

Properties of a material

What you should know

By the end of this section you should be able to:

- Describe bonding as a continuum between the ionic, covalent and metallic models.
- Use bonding models to explain the properties of a material.

In this topic so far, we have looked at ionic, covalent and metallic models as a way to categorise the arrangement of atoms based on electrostatic interactions. Throughout the twentieth century, many chemists used the models of bonding to categorise substances and link the three bonding models together. **Interactive 1** shows some of the examples that were developed. Note that many of the models, terms and symbols used no longer align with those used today. As you scroll and look through the historical methods of categorisation, identify similarities and differences with what you have learned about ionic, covalent and metallic models of bonding.

Interactive 1. Bonding Models and Categorisation of Substances Based on Structure.

In this section, we will discuss the utility and limitations of these three bonding models in explaining the properties of a material. As we think about how to explain the properties of a material consider the following:

- How are the three bonding models useful?
- Are there any limitations to these three bonding models?
- Do you think all substances can be neatly classified into one of the three bonding models?

Chemical bonding

So far, we have classified chemical bonding based on the electrostatic interactions between charged particles:

- Ionic – between oppositely charged ions.
- Covalent – between a shared pair of electrons and positively charged nuclei.
- Metallic – between cations in a lattice and delocalised electrons.

What type of elements (metal/non-metal) would you select that combine to form each type of bonding? We can use a bonding triangle to classify the type of chemical bonding that occurs based on the type of elements that make up the substance, as shown in **Figure 1**.

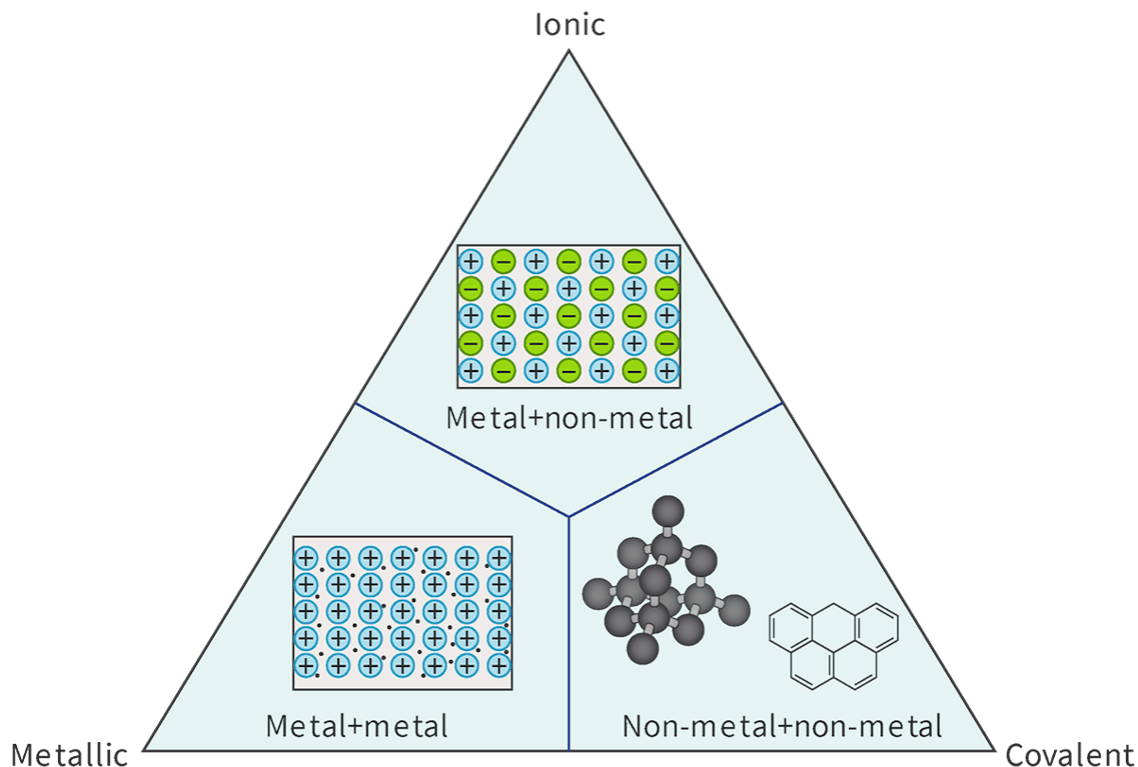


Figure 1. Triangle representing the three types of bonding.

 More information for figure 1

The image is a triangle diagram representing three types of chemical bonding based on the elements involved. The triangle is divided into three zones:

1. **Metal + Metal** (Bottom left corner): This section represents metallic bonding, depicted with a grid of positive symbols, indicating metal ions in a sea of electrons.
2. **Metal + Non-Metal** (Top section): This part of the triangle shows ionic bonding, illustrated with a grid-like pattern containing alternating positive and negative symbols, suggesting the attraction between metal cations and non-metal anions.
3. **Non-Metal + Non-Metal** (Bottom right corner): This zone corresponds to covalent bonding. It includes molecular structures, such as a cluster of spheres and a hexagonal shape, symbolizing shared electron pairs between non-metal atoms.

The triangle serves to categorize chemical bonds by placing them at their respective vertices, based on the elements' metallic or non-metallic nature.

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This diagram allows classification of a bond between two elements by placing it on a vertex on the triangle, depending on whether the elements are metals or non-metals. If the elements making up a chemical bond are both metals, the bonding is classified as metallic. If the chemical bond is made up of a metal and non-metal, the bonding is classified as ionic. Finally, if the bond is made up of two non-metals, the bonding is classified as covalent. A more complex version of this triangle can be found in [section 17 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/triangular-bonding-diagram-van-arkelketelaar-id-45119/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/triangular-bonding-diagram-van-arkelketelaar-id-45119/) of the DP chemistry data booklet. We will look more extensively at the features of this triangle in [section S2.4.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bonding-triangle-id-43843/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bonding-triangle-id-43843/).

Notice that the three bonding models are placed together in one triangle instead of separating them entirely. What would be the purpose of putting them together? What do you think happens along the boundaries between two bond types? Is it possible for a bond to have characteristics that fit in more than one category?

International Mindedness

The boundary lines that divide ionic, covalent and metallic bonding can be viewed in parallel to the border boundaries that divide nations, territories, districts, states and provinces. Border communities often exhibit similarities in culture, language, food, etc. on both sides of the boundary.

Bonding as a continuum

In the triangle model in **Figure 1**, the edges of the triangle would represent bond types of substances that fall somewhere between two bond types. Can you think of an example of a bond type that falls somewhere in between covalent and ionic bonding? **Figure 2** shows an example of a bonding continuum that exists along the edge of the triangle between covalent and ionic bonding. This bonding continuum shows the transition between pure covalent bonding (e.g. Cl_2) to ionic bonding (e.g. NaCl), with polar covalent bonding lying in between (e.g. HCl).

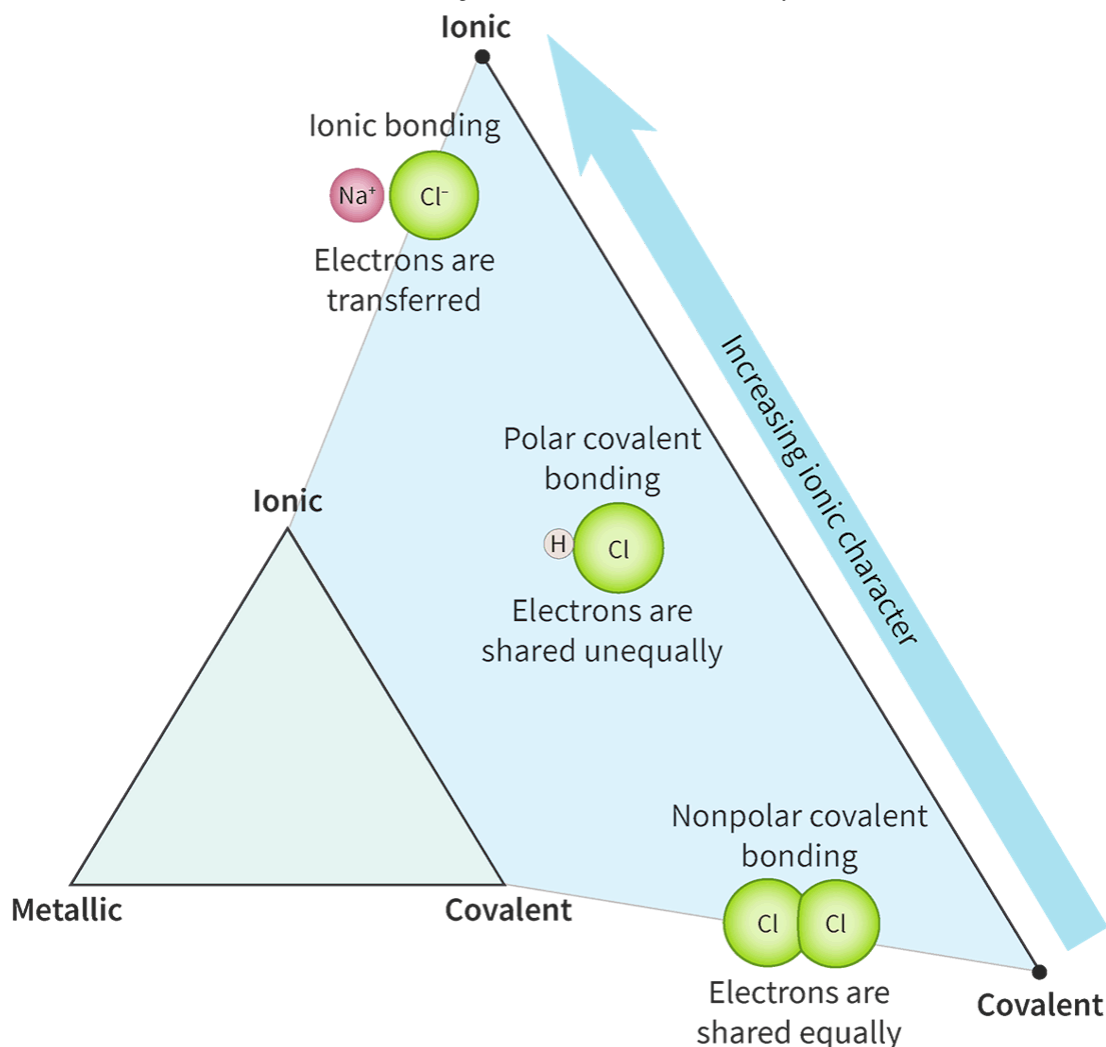


Figure 2. Bonding continuum between covalent and ionic bonding.

 More information for figure 2

This diagram illustrates the bonding continuum between covalent and ionic bonding within a triangular model. The left side features Na^+ and Cl^- ions labeled as ionic bonding, with a note stating that electrons are transferred. The right side has a gradient labeled "Increasing ionic character," showing polar covalent bonding between H and Cl, where electrons are shared unequally. At the bottom, nonpolar covalent bonding is depicted with two Cl atoms. This triangular representation links ionic, polar covalent, and nonpolar covalent bonds along the continuum.

[Generated by AI]

A bonding continuum similarly exists from metallic to ionic bonding, and from metallic to covalent bonding. This means that bonding is best described as a continuum between the ionic, covalent and metallic models rather than discrete categories.

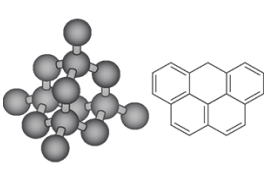
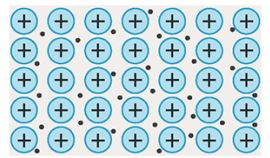
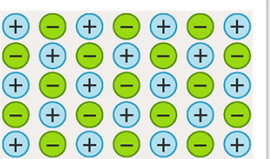
Nature of Science

Aspect: Models

It is important to remember that while models can be useful all models have limitations. In the triangle model used to represent bonding in this section, it is a useful model because it allows bonding to be viewed as a continuum rather than discrete. However, the limitation of this particular model is that it does not help you identify which bonds would lie on the extremes (vertices of the triangle) and which bonds would lie between two types of bonding (centre of an edge on triangle). Highlighting the limitations of a model allows for new models to be developed that can more accurately and completely represent natural phenomena.

Properties of a material using bonding models

The physical characteristics of materials used in our everyday lives are closely related to the structure and bonding of the elements. Try out **Interactive 2** below to see whether you can recall the bonding and structure of each of the three types of chemical bonding. Drag the descriptions of the bonding and structure to match the images in the table.

	Metallic	Covalent	Ionic
Bonding description			
Description of structure			
Structural diagram		Small or large molecules, covalent network structures	Electrostatic attraction between positive metal ions and delocalised valence electrons
	Lattice structure	Electrostatic attraction between oppositely charged ions	
	Electrostatic attraction between shared pairs of electrons and positively charged nuclei		Lattice structure
✓ Check			

Interactive 2. Bonding and Structure Descriptions and Images.

 More information for interactive 2

An interactive table with missing fields where users need to drag and drop the bonding descriptions, descriptions of structure, and structural diagrams of the metallic, covalent, and ionic bonds.

The drag-and-drop options are as follows:

A structure on the left shows a 3D model representing a diamond structure with a tetragonal arrangement. Each carbon atom is bonded to four other carbon atoms. To the right, a chemical structure features five fused benzene rings, three on the top and two on the bottom.

Lattice structure

Electrostatic attraction between shared pairs of electrons and positively charged nuclei

Small or large molecules, covalent network structures

Electrostatic attraction between oppositely charged ions

A structure illustrates an alternating arrangement of positive and negative ions

Electrostatic attraction between positive metal ions and delocalised valence electrons

A structure of positively charged metal ions surrounded by a sea of electrons

Lattice structure

This interactivity helps users to identify and differentiate between metallic, covalent, and ionic bonds.

Read below for the solution:

Bonding description:

Metallic: Electrostatic attraction between positive metal ions and delocalised valence electrons.

Covalent: Electrostatic attraction between shared pairs of electrons and positively charged nuclei.

Ionic: Electrostatic attraction between oppositely charged ions.

Description of structure:

Metallic: Lattice structure.

Covalent: Small or large molecules, covalent network structures.

Ionic: Lattice structure.

Structure diagram:

Metallic: A structure of positively charged metal ions surrounded by a sea of electrons.

Covalent: A structure on the left shows a 3D model representing a diamond structure with a tetragonal arrangement. Each carbon atom is bonded to four other carbon atoms. To the right, a chemical structure features five fused benzene rings, three on the top and two on the bottom.

Ionic: A structure illustrates an alternating arrangement of positive and negative ions.

The three bonding models can be used to explain the properties of a material.

What are some physical properties that we have looked at so far? Which ones are unique to ionic bonding? Covalent bonding? Metallic bonding? Which type of bonding is the most difficult to assign specific physical properties to?

Covalent bonding is usually the most difficult to assign specific physical properties because these properties depend on intermolecular interactions and whether or not the substance is a small molecule, large molecule or covalent network solid.

Materials chemists tend to classify materials according to the structure and bonding of the elements making up a substance; however, many scientists and engineers choose to classify materials based on their physical characteristics. Many of these physical characteristics are the physical properties we looked at in section S2.1.3b (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-properties-of-ionic-compounds-id-43559/>) for ionic compounds, section S2.2.7 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/>) for covalent network structures and section S2.3.1b (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-properties-and-application-of-metals-id-43563/>) for metals. **Figure 3** is a summary of some of the common physical properties used to classify materials.

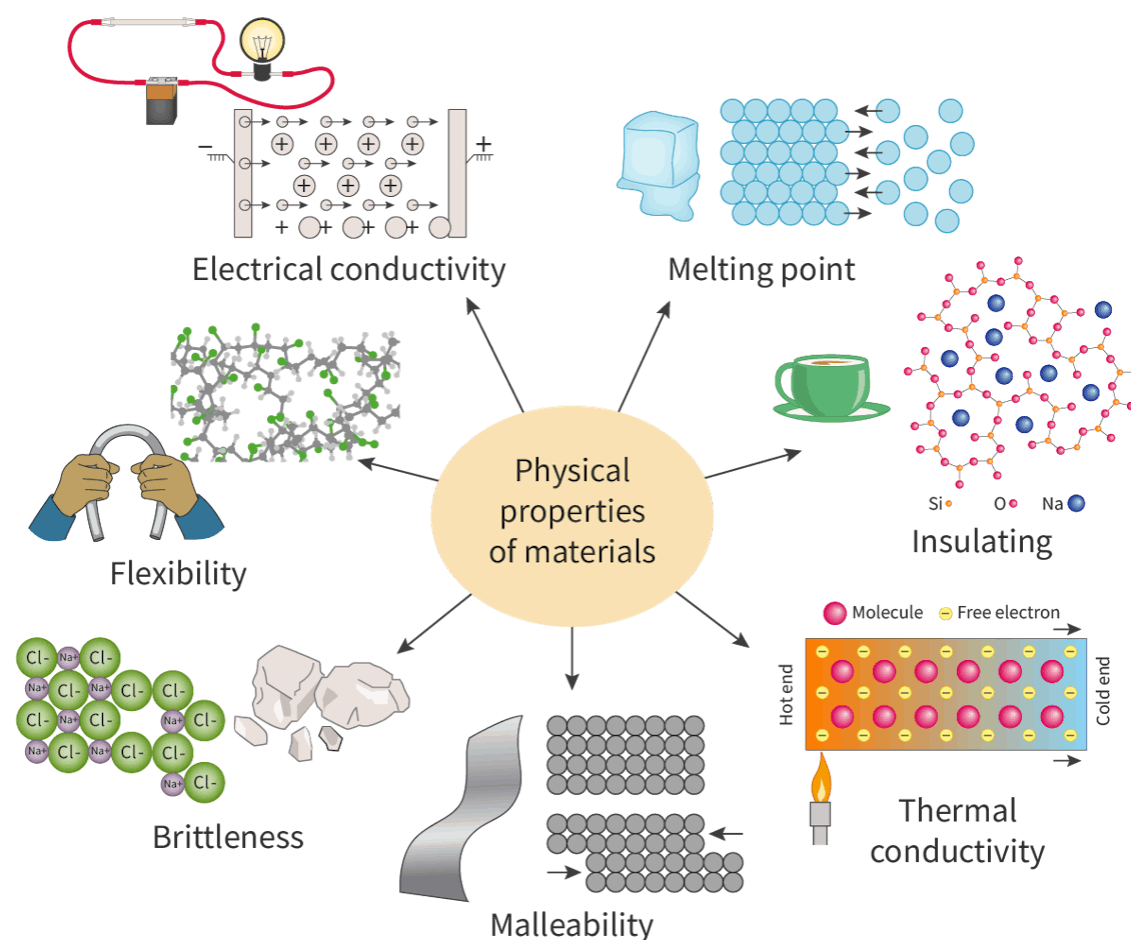


Figure 3. Physical properties of some materials.

 More information for figure 3

The diagram illustrates various physical properties of materials with examples. At the center is a beige circle labeled "Physical properties of materials," with lines extending to depict different properties.

1. **Electrical Conductivity:** Illustrated with an image of an electric circuit showing a battery connected to a light bulb via conductors, demonstrating current flow.

2. Melting Point: Depicted by ice and a grid of particles transitioning from solid to liquid state.
3. Flexibility: Shown using hands bending a bar, with a molecular structure illustrating flexible bonds.
4. Brittleness: Represented by a material breaking along its crystalline lines, with Na^+ and Cl^- ions depicted.
5. Malleability: Illustrated by bending a sheet of material, depicted by particles rearranging under pressure.
6. Insulating: Shown with a ceramic cup and a molecular structure where Si, O, and Na are connected, highlighting resistance to heat or electricity flow.
7. Thermal Conductivity: Depicted by a bar with a hot and cold end, showing molecules and free electrons transferring heat between ends.

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Do you notice any physical properties listed that were not discussed when we looked at ionic, covalent and metallic bonding? Flexibility is the ability of a material to be bent without breaking. This property is similar to malleability. Flexible materials can return to their original shape after experiencing a compression whereas malleable materials remain bent. Insulating is when a material does not permit thermal energy or an electric current to flow freely. This is usually due to the absence of freely moving particles.

When designing a material you would need to ask yourself what properties you would want from the material. The questions below are examples of those that might be asked by someone designing a material object. For each of the questions, consider what type of bonding would have that property.

- Does it need a high melting point?
- Does it need to be electrically conductive in the solid state?
- Does it need to be hard?
- Will it shatter if you drop it?

- Is it shiny?
- Does it bend easily?
- Do you need it to insulate or conduct heat (thermal energy)?

We know that the physical properties of a material are dependent on the chemical structure and bonding that make it up. In this activity you will use your experiences with everyday objects to predict the bond type you would expect, followed by reviewing explanations for the common physical properties of ionic, metallic and covalent substances.

Activity

- **IB learner profile attribute:** Communicators
- **Approaches to learning:** Social skills — Clearly communicating complex ideas in response to open-ended questions
- **Time required to complete activity:** 30—40 minutes
- **Activity type:** individual activity to start then pair activity to follow

Copy and complete the following table for the objects listed. List all the physical properties that you think the object would have (2—5 properties per object). Looking at your list of physical properties, assign the type of bonding you would expect the elements making up this object to have. For each bonding type assigned, provide a percentage to show how confident you are in your estimated bonding type (e.g. if you were deciding between two bonding types you might list 50% confident, if you were really certain about the bonding type you might list 100% confident).

Object	Physical properties of object	Estimated bonding type	P
Silver spoon 			

Object	Physical properties of object	Estimated bonding type	P
Eggshell 			
Latex balloon 			
Brick 			

Object	Physical properties of object	Estimated bonding type	P
Pencil lead 			
Stainless steel frying pan 			

With a partner, compare your tables.

Did you and your partner have any similarities in a bonding type you were both **confident** in assigning? If so, which bonding type? Discuss reasons why you found it **easy** to assign this bonding type.

Did you and your partner have any similarities in a bonding type you both **lack confidence** in assigning? If so, which bonding type? Discuss reasons why you found it **difficult** to assign this bonding type.

In this next part of the activity you will explain, orally, to your partner why ionic compounds or metallic substances have the physical properties that they do. One partner will explain the physical properties for ionic compounds while the other partner will explain metallic substances.

Ionic compounds are...	Metallic substances are...
Brittle	Malleable
Insulating	Conductive

Ionic compounds are...	Metallic s
Solids at room temperature	Solids at r

Covalent materials are more difficult to assign physical properties to because the properties can vary depending on how atoms are arranged, whether there are intermolecular forces or covalent bonds holding the solid together, and whether there are free electrons within the structure.

With your partner, brainstorm an example of a covalent solid for each row that would have the physical property stated:

Physical property	Example o
Electrically conductive	
High melting point	
Hard	
Soft	